Clustering based on W2V

- Train a Word2Vec Model
  - create a text corpus (e.g., Wikipedia dump, or works from the Project Gutenberg, such as https://www.gutenberg.org/ebooks/).
  - preprocess the text through tokenization, lowercasing, and stopwords and punctuation removal.

Clustering based on W2V

- train a Word2Vec Model (using `gensim.models.Word2Vec`):
  - choose hyperparameters: vector_size, window, min_count, sg
  - save the trained model for later use
- explore the Model:
  - use `.most_similar()` to explore semantic neighborhoods
  - visualize some vectors using PCA or t-SNE

Clustering based on W2V

- Words Clustering
  - Choose a list of words
  - Cluster them using scikit-learn's KMeans
  - Visualize the resulting clusters using t-SNE or PCA

